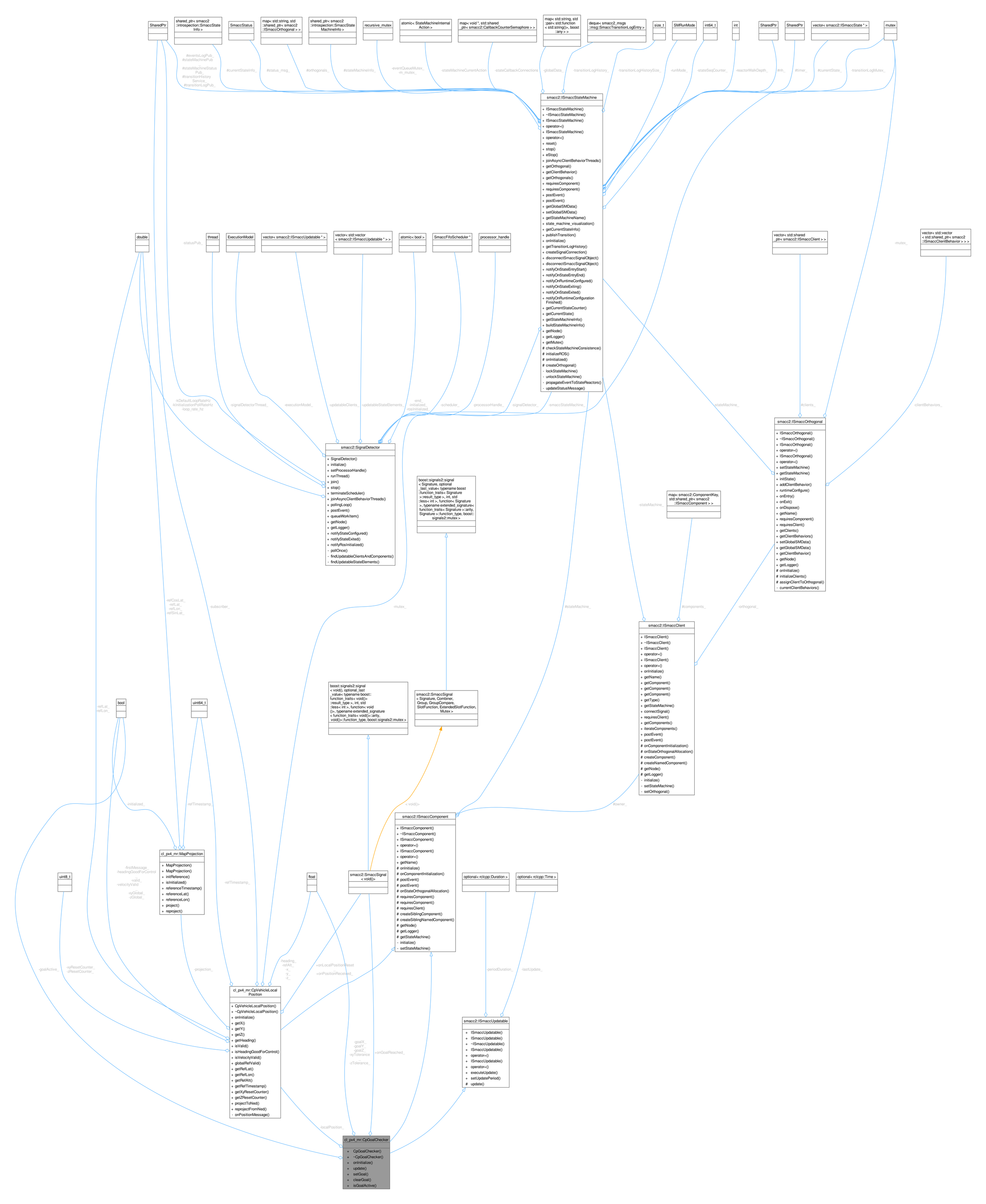
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
int
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
mutex
#eventsLogPub_
#stateMachinePub
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
-reactorWalkDepth_
#nh_
#timer_
#currentState_
-transitionLogMutex_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
double
-statusPub_
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< bool >
SmaccFifoScheduler
processor_handle
vector< std::shared _ptr< smacc2::ISmaccClient > >
-mutex_
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetector_
-smaccStateMachine_
-stateMachine_
#clients_
-clientBehaviors_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
-refCosLat_
-refLat_
-refLon_
-refSinLat_
-subscriber_
-mutex_
#stateMachine_
#components_
-orthogonal_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
bool
uint64_t
-refLat_
-refLon_
-initialized_
-refTimestamp_
< void()>
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
cl_px4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
-firstMessage_
-headingGoodForControl
-valid_
-velocityValid
-xyGlobal_
-zGlobal_
uint8_t
float
smacc2::SmaccSignal < void()>
optional< rclcpp::Duration >
optional< rclcpp::Time >
-goalActive_
-xyResetCounter_
-zResetCounter_
-projection_
-heading_
-refAlt_
-x_
-y_
-z_
+onLocalPositionReset
+onPositionReceived_
-periodDuration_
-lastUpdate_
cl_px4_mr::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ ISmaccUpdatable()
+ operator=()
+ ISmaccUpdatable()
+ operator=()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-goalX_
-goalY_
-goalZ_
-xyTolerance_
-zTolerance_
+onGoalReached_
-localPosition_
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()